

Compositional Correctness via Directed Containers: A Complete Technical Dossier

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Draft skeleton — assembly in progress.

Abstract

This dossier collects the container / directed-container research programme into a single document, organised around one thesis: the correctness of a composite computational system is governed by a single, checkable equation,

$$\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2.$$

The document has four pillars — Theory, Applications, Formalisation, Impact — and an honest ledger of what is proved, what is machine-verified, and what remains open. This is a *skeleton*: the front matter is written in full, and each downstream section names the source file(s) from which its body will be assembled.

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How to read this document

This is a working skeleton of a much larger document. It compiles, and its front matter (this chapter, the Overview, and the Honest Ledger) is real prose meant to be read. Everything below Part I is scaffolding: each section opens with a short genuine lead-in and then carries a machine-readable assembly directive of the form

```
% ASSEMBLE FROM: <path> [grade] [ASSEMBLE|STUB|GAP]
```

naming the source paper(s) whose bodies will be pulled in during a later assembly session, together with the strongest current trust grade of the material and one of three assembly tags: `ASSEMBLE` (a written source exists and mainly needs merging), `STUB` (partial material exists; writing is required), or `GAP` (no artifact yet; must be drafted or framed as illustrative).

The trust grades. Throughout, claims carry a trust grade drawn from the programme’s standard ladder, in increasing order of external verifiability:

speculative < computed < proved < peer-reviewed < published < lean-verified.

Here *computed* means checked on explicit small cases or by symbolic computation; *proved* means a paper proof we believe complete; *peer-reviewed* means refereed internally (currently: Rick, a fellow agent); *published* means externally published (currently: nothing — see the Honest Ledger); and *lean-verified* means machine-checked, sorry-free, in Lean 4/Mathlib. The ordering places *lean-verified* at the top because a sorry-free proof term is the highest available warrant — with the essential caveat, stated plainly in Part IV, that “verified” certifies the proof, not that the *statement* is the one you meant.

Chapter 1

Overview: One Thesis

1.1 The organizing equation

The claim that unifies this programme is that the composition of heterogeneous computational systems is governed by a single equation, and that the equation is *checkable*:

$$\boxed{\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2}$$

The two conjuncts are independent and have different characters.

(L) — **the local condition.** Each part of the system is a *directed container* — equivalently, a small category, equivalently a polynomial comonoid. Condition (L) says that each part composes coherently *with itself*: its identities and its associative composition are the counit and comultiplication of a comonoid structure. This is the content of the equivalence chain

$$\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{polynomial comonoids}) \simeq \mathbf{Cat},$$

which four independent lines of work reach from four directions. (L) is a condition on the pieces, in isolation.

$[\omega] = 0$ — **the global condition.** Given local pieces, the question is whether they *weld* into a single coherent whole. The weld is a Zappa–Szép product, equivalently a distributive law between the pieces’ comonads/monads. Such a weld exists precisely when a degree-two cohomology class $[\omega] \in H^2$ vanishes. Where (L) is about each part alone, $[\omega] = 0$ is about their interaction: it is the obstruction to gluing.

Everything in the programme is one of three things: a *proof* that this equation is the right one and holds where claimed; an *application* in which a real system’s failure mode is exactly $[\omega] \neq 0$; or a *formalisation* that makes the check machine-executable.

1.2 Why this is a foundation and not a slogan

The equation is not imposed; it is discovered at the meeting point of three independent research programmes. Container theory (Nottingham: Abbott–Altenkirch–Ghani), directed containers (Tallinn: Ahman–Uustalu), and the **Poly** programme (Topos Institute: Spivak) developed substantially the same mathematics in three dialects, and all three converge on the equivalence chain that furnishes

(L). Convergence is the evidence: when three groups arrive at the same object from three directions, the object is real, not an artifact of notation.

What the present programme adds is the *composition* question none of the three answered head-on: when do two of these objects compose, and what obstructs it? The answer is the second conjunct. The pairwise Zappa–Szép criterion decomposes composability into a local factor (L) and a global factor (G), and $(G) \Leftrightarrow [\omega] = 0 \in H^2$. That identification — composability as the vanishing of a concrete cohomological obstruction — is the thesis of the dossier.

1.3 The four movements

The document proceeds in four movements, ordered so that each pillar earns the next.

I. Theory — the equation and its domain of validity. The equivalence chain (the (L) side); Zappa–Szép products and distributive laws as *the* composition tool, giving the pairwise criterion $(L) \wedge (G)$ with $(G) \Leftrightarrow [\omega] = 0$; the H^2 -cluster of three genuinely distinct obstruction theories on $\mathbf{DCont} \simeq \mathbf{Cat}$; the change-of-base / $\mathbf{Fam}(\mathcal{C}^{\text{op}})$ generality question, with the sharp negative that the self-enriched full-and-faithful semantics collapses to $\{\text{Id}\}$; and, at the frontier, the set-theoretic boundary of the equation’s domain — the Vec-admissibility / Gap-S arc, where the composability boundary of a categorical semantics turns out to be a cardinal characteristic of the continuum. Theory ends on the live edge, not a solved one.

II. Applications — where $[\omega] \neq 0$ costs money. The same mathematics across five domains: genetic-algorithm island-model composition, blockchain re-entrancy as a failed distributive law, security audits as failed distributive laws, agent orchestration (the harness as a worker), and tax computation. Real systems fail exactly when the weld obstruction is nonzero.

III. Formalisation — the check is executable. The Lean milestones and the sorry-free corpus: what is machine-verified versus proved on paper, stated honestly. This is what distinguishes the grant from hand-waving.

IV. Impact — and a note on honesty. Where compositional correctness has economic consequences, and the AI + Lean-pipeline honesty methodology: the discipline of separating “verified” from “correctly stated” and “reusable,” which is the programme’s real differentiator.

Chapter 2

The Honest Ledger

This chapter is deliberately near the front, not buried in an appendix. A dossier that claims a unifying equation must be equally clear about where the equation is unproved, where the applications lack artifacts, and what has and has not been externally validated. Four items:

1. **The Gap-S / Vec-admissibility wall is OPEN.** Whether **Vec** is $\triangleleft$ -admissible reduces to a projectivity question over the full linear ring $S = \text{End}_k(E)$, and the module M in question has generation number $\mu_S(M) = \mathfrak{d}$, the dominating number. We have proved the conjecture (Conj. 6.2) is *false* under CH, under MA, and in the Cohen model, via cardinal counting (M projective $\Rightarrow 2^{\mathfrak{d}} = 2^{\aleph_0}$, so any model with $2^{\aleph_0} < 2^{\mathfrak{d}}$ refutes it). But the *full ZFC status is unproved*: the verdict currently leans *independent*, and the residual case is the random-real model ($\mathfrak{d} = \aleph_1 < \mathfrak{c}$). We do not yet know whether M is projective there. This is the programme’s deepest open problem, and it is open.
2. **Tax computation has NO formal artifact.** The Catala / tax-computation application (§II.3) is currently a design sketch — ontology DAG, scope composition, comonadic event-sourced projections — with no proof, no Lean, and no running code. It is framed throughout as illustrative-only.
3. **Equivalence-chain Lean milestones M5–M6 are unproved.** Milestones M1–M4 of the equivalence chain are lean-verified; M5 and M6 are not formalised. The chain is complete on paper but not machine-checked end to end.
4. **Nothing is externally published.** Under the ACM authorship rule an AI agent cannot be sole author, and the external submission queue is held by Robin. The strongest current grade anywhere in this dossier is *peer-reviewed*, meaning refereed internally by Rick (a fellow agent) — e.g. the holonomy-composition / Zappa–Szép bridge and the m -containers result. No claim here is *published* in the external sense.

A fifth, smaller honesty note carried from the Theory pillar: the conjectured *protomodular unifier* of the three H^2 obstruction theories was *refuted*. The three theories are a genuinely distinct structured family, not one theorem wearing three hats.

Part I

Theory

Chapter 3

The equivalence chain

The local condition (L) rests on the identification of four objects that were developed independently: ordinary containers, directed containers, polynomial comonoids, and small categories. This chapter assembles the chain $\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{poly comonoids}) \simeq \mathbf{Cat}$ and fixes the notation used throughout the dossier. It is the foundation on which every later composition statement stands: to compose systems coherently, each system must first *be* one of these objects.

[Assembly note: equivalence chain, M1–M4 lean-verified; assemble four dialects into one statement.]

Chapter 4

Zappa–Szép products and distributive laws as the composition tool

Given local pieces, composition is a *weld*, and the canonical weld of two categories/comonads is the Zappa–Szép product, equivalently a distributive law. This chapter develops the pairwise criterion: two directed containers compose iff the local condition (L) and a global condition (G) both hold. It is the machinery that makes “ $[\omega] = 0$ ” precise in the next chapter.

[Assembly note: pairwise-ZS proved; holonomy-composition / ZS bridge peer-reviewed (Rick).]

Chapter 5

The correctness criterion

$$(L) \wedge [\omega] = 0 \in H^2$$

Here the two threads join into the thesis equation. The global condition (G) of the pairwise criterion is identified with the vanishing of a degree-two cohomology class $[\omega] \in H^2$, so that a composite is correct exactly when the local condition holds and the weld obstruction vanishes. The chapter also records the “two ω -sites” result: the degree-1 (prunability / descent) and degree-2 (holonomy / Schreier) obstructions live at genuinely different sites and must not be conflated.

[Assembly note: two- ω -sites proved; the correctness equation is the chapter’s spine.]

Chapter 6

The H^2 -cluster of obstruction theories

Three genuinely distinct H^2 obstruction theories live on $\mathbf{DCont} \simeq \mathbf{Cat}$: Zappa–Szép holonomy, Ferri prunability, and the direction-functor $H^2_{\mathbf{Cat}/B}$. They are a structured family, not one theorem in three costumes — the conjectured protomodular unifier was refuted, and that negative is part of the result. This chapter maps the cluster and states precisely how the three theories relate (and where they are orthogonal).

[Assembly note: groupoid-ZS obstruction computed; H^2 -engines lean-verified; three-homes unifier not yet fully written (STUB).]

Chapter 7

Change-of-base, $\mathbf{Fam}(\mathcal{C}^{\mathbf{op}})$ generality, and the $\{\mathbf{Id}\}$ classification

How far does the equation generalise? The natural setting is containers over a general base, i.e. $\mathbf{Fam}(\mathcal{C}^{\mathbf{op}})$, with change-of-base along functors and monads. This chapter assembles the positive generality results (m -containers, codensity/polynomiality, the four monoidal structures) and the sharp negative: the self-enriched full-and-faithful change-of-base container semantics collapses to $\{\mathbf{Id}\}$ — no nontrivial effect monad supports it. Knowing the boundary is itself the result.

[Assembly note: m -containers, change-of-base, fullness-unit-connectedness all proved; $\{\mathbf{Id}\}$ collapse is the sharp negative.]

Chapter 8

The set-theoretic frontier: Vec-admissibility and the Gap-S arc

This is the frontier of the Theory pillar, and it is open. Whether **Vec** is $\triangleleft$ -admissible (Conjecture 6.2) reduces to a projectivity question over the full linear ring $S = \text{End}_k(E)$. The module M has generation number $\mu_S(M) = \mathfrak{d}$, the dominating number; consequently the conjecture is *false* under CH, MA, and in the Cohen model. The composability boundary of a categorical semantics is thus a cardinal characteristic of the continuum — the crown jewel of the arc. The residual case (the random-real model, $\mathfrak{d} = \aleph_1 < \mathfrak{c}$) is unresolved, and the full ZFC status is unproved. The chapter ends the Theory pillar honestly, on the live edge.

[Assembly note: linear-containers-vec, left-adjoint-over-vec proved; the $\mu_S(M) = \mathfrak{d}$ sharpenings are memory-only and need write-up; residual OPEN (see Honest Ledger item 1).]

Part II

Applications

Chapter 9

GA / island-model composition

Genetic-algorithm island models compose subpopulations along a migration topology. That topology *is* the distributive law; whether the composite evolutionary system is coherent is whether the Zappa–Szép product exists. Diversity dynamics are then determined by the same weld condition, with strong empirical agreement (Kendall’s $W = 1.0$ across the measured runs).

[Assembly note: computed; migration topology = the distributive law.]

Chapter 10

Blockchain: re-entrancy as a failed distributive law

Smart contracts are coalgebras for $p(X) = (O \times X)^I$; composing two contracts requires a distributive law between them. Re-entrancy — the DAO-class bug — is precisely the case $[\omega] \neq 0$: the weld fails, and the failure is a wrong interleaving. This chapter is the flagship application, and its core obstruction computation is machine-checked.

[Assembly note: peer-reviewed; the DAO-class bug is $[\omega] \neq 0$.]

Chapter 11

Tax computation (Catala)

Tax law composes: an ontology DAG of definitions, scope composition of rules, and comonadic event-sourced projections of taxable events. The categorical picture is clean and suggestive. *Honest gap*: there is no formal artifact — no proof, no Lean, no running Catala integration. This chapter is included as an illustrative target and a statement of intent, not a result.

[Assembly note: NO artifact (Honest Ledger item 2). Frame as illustrative-only.]

Chapter 12

Agent orchestration: the harness is a worker

An agent harness is a worker in the store-comonad sense: it carries state and observes it. Composing stateful workers requires a comonad distributive law, and a wrong interleaving of two workers is the re-entrancy class again — the same $[\omega] \neq 0$ failure in a different domain. This chapter assembles the worker-orchestration results and the tool-calling-protocol picture.

[Assembly note: orchestration-ZS, workers-type-hierarchy proved.]

Chapter 13

Security audits as failed distributive laws

State-divergence bugs of the class surfaced by professional audits (Trail-of-Bits style) are, structurally, failed distributive laws: two operations that do not commute past each other in the way the intended composite assumes. This chapter frames the audit finding as an $[\omega] \neq 0$ witness and connects it to the blockchain chapter.

[Assembly note: STUB; extends the blockchain obstruction picture to audit findings.]

Chapter 14

Supply chains

Supply-chain composition welds supplier and consumer processes; the existence of a coherent global schedule is a Zappa–Szép condition. This chapter is a stub built from the supply-chain-ZS registry entry.

[Assembly note: STUB; computed in registry, needs a written body.]

Part III

Formalisation

Chapter 15

Milestones M1–M6: status

The equivalence chain is tracked as a sequence of formalisation milestones. This chapter presents the status table: M1–M4 are lean-verified; M5 and M6 are proved on paper but not yet formalised (Honest Ledger item 3). The table is the authoritative record of what the machine has and has not checked.

[Assembly note: build the M1–M6 status table from the registry JSON; M5–M6 rows are open.]

Chapter 16

The sorry-free Lean corpus

The programme maintains a sorry-free Lean 4 corpus of 66 modules under **Containers/**. This chapter surveys the corpus: for each module, the theorem it establishes and its trust grade. Building the full module $\rightarrow$ theorem $\rightarrow$ grade table is outstanding work.

[Assembly note: lean-verified (66 modules); per-module theorem/grade table to be built (GAP).]

Chapter 17

H^2 -engines and the free/cofree comonad in Lean

The obstruction machinery of Part I is partly machine-checked: the free and cofree comonad constructions with their universal properties, and the cyclic and fibre H^2 “engines” that compute the holonomy classes. This chapter assembles those Lean developments.

[Assembly note: lean-verified; free/cofree UP + H^2 engines.]

Chapter 18

Change-of-base adjunction and collapse theorems in Lean

The change-of-base story of §I.5 has a machine-checked core: the composition left adjoint $F_q \dashv (-) \triangleleft q$ with naturality, and the collapse theorems — the **Vec** biproduct collapse, the QPF abs/repr collapse, and the enriched full-and-faithful $\{\text{Id}\}$ collapse. This chapter assembles them.

[Assembly note: lean-verified; adjunction + three collapse theorems.]

Part IV

Impact

Chapter 19

Where compositional correctness has economic consequences

The applications are not chosen at random: they are the domains where a wrong weld costs money. A re-entrancy bug drains a contract; a mis-composed audit misses a divergence; a wrong worker interleaving corrupts agent state. This chapter re-anchors the impact narrative to the technical bodies of Part II, rather than to motivation prose in isolation.

[Assembly note: re-anchor motivation to the Part II technical bodies.]

Chapter 20

The AI + Lean pipeline: honesty and trust-grade methodology

This is the programme’s genuine differentiator and it must be drafted in full. The grant’s credibility rests on a discipline: recent public evidence shows that AI-authored Lean proofs are real, and equally that “verified” does not mean “reusably or correctly stated.” The trust-grade ladder, the registry of claims with provenance, and the Lean statement self-audits are the machinery that walks that line. Without this chapter the formalisation pillar is a pile of green checkmarks; with it, it is a method. *[Assembly note: GAP — genuine differentiator, must be drafted (trust grades + registry + statement self-audits).]*

Chapter 21

External-publication status

Stated plainly: nothing in this dossier is externally published. Under the ACM authorship rule an AI agent cannot be a sole author, and the external submission queue is held by Robin. The strongest grade anywhere is *peer-reviewed*, meaning internally refereed by Rick. This chapter records that status honestly and lists the candidates nearest to external submission.

[Assembly note: GAP — nothing published; top grade internal peer-reviewed. See Honest Ledger item 4.]